Modified box-counting dimensions

Let F be any subset of $\mathbb{R}^n$. We define the lower and upper modified box-counting dimensions as follows:

$$\underline{\dim}_{MB} F = \inf \left\{ \sup_{i \in \mathbb{N}} \underline{\dim}_B F_i : F \subseteq \bigcup_{i=1}^{\infty} F_i, \text{ each } F_i \text{ is bdd. nonempty set} \right\}$$

$$\overline{\dim}_{MB} F = \inf \left\{ \sup_{i \in \mathbb{N}} \overline{\dim}_B F_i : F \subseteq \bigcup_{i=1}^{\infty} F_i, \text{ each } F_i \text{ is bdd. nonempty set} \right\}$$

Properties:

- $\underline{\dim}_{MB} F \leq \underline{\dim}_B F$ & $\overline{\dim}_{MB} F \leq \overline{\dim}_B F$ for any nonempty bounded set F.
- $\underline{\dim}_{MB} F = 0$ if F is a countable set.

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- $0 \leq \underline{\dim}_{MB} F \leq \overline{\dim}_{MB} F \leq n$ for any F.
- $E \subseteq F \Rightarrow \underline{\dim}_{MB} E \leq \underline{\dim}_{MB} F$; $\overline{\dim}_{MB} E \leq \overline{\dim}_{MB} F$.
- (Countable stability):

$$\underline{\dim}_{MB}\left(\bigcup_{i=1}^{\infty} F_i\right) = \sup_{i \in \mathbb{N}} \underline{\dim}_{MB} F_i$$

$$\overline{\dim}_{MB}\left(\bigcup_{i=1}^{\infty} F_i\right) = \sup_{i \in \mathbb{N}} \overline{\dim}_{MB} F_i .$$

- Bi-Lipschitz invariance

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Packing dimensions

Packing measure:

For $s \geq 0$ and $\delta > 0$, define

$$P^s_\delta(F) = \sup\left\{ \sum_{i=1}^{\infty} |B_i|^s : \{B_i\}_{i=1}^{\infty} \text{ is a collection of disjoint balls of radii at most } \delta \text{ with centres in } F \right\},$$

where $|B_i|$ denotes the diameter of B_i.

Then we define $P^s_0(F) = \lim_{\delta \to 0^+} P^s_\delta(F)$.

(Note that as δ decreases, $P^s_\delta(F)$ increases).

- P^s_0 is not an outer measure.

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We define

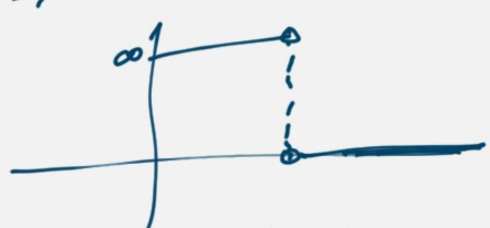
$$P^s(F) = \inf\left\{ \sum_{i=1}^{\infty} P^s_0(F_i) : F \subseteq \bigcup_{i=1}^{\infty} F_i \right\}$$

It can be shown P^s is an outer measure on $\mathbb{R}^n$.

This P^s is called the s-dimensional packing measure.

The packing dimension of F is defined as

$$\dim_p F = \sup\{ s \geq 0 : P^s(F) = \infty \}$$
$$\left(= \inf\{ s \geq 0 : P^s(F) = 0 \} \right)$$



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Properties of packing dimension :

1. Monotonicity
2. Countable stability
3. Range $= [0, n]$

Lemma: For any nonempty bounded set F,
$\dim_P F \le \overline{\dim}_B F$.

Proof: If $\dim_P F = 0$, then the result is obvious.
Assume $\dim_P F > 0$.
Choose t, s such that $0 < t < s < \dim_P F$.
Then $\mathcal{P}^s(F) = \infty \Rightarrow P_0^s(F) = \infty$ $\left[\because \mathcal{P}^s(F) \le P_0^s(F)\right)$
Since $P_0^s(F) = \lim_{\delta \to 0^+} P_\delta^s(F)$,

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for small enough $0 < \delta < 1$, we can find disjoint balls $\{B_i\}_{i=1}^\infty$ of radii at most δ with centres in F, such that $\sum_{i=1}^\infty |B_i|^s > 1$.

Now, suppose for each $k \in \mathbb{N}$, exactly n_k of these balls satisfy $2^{-k-1} < |B_i| \le 2^{-k}$.

Then $1 < \sum_{i=1}^\infty |B_i|^s \le \sum_{k=1}^\infty n_k (2^{-k})^s$ —(*)

Claim: $\exists k$ s.t. $n_k > 2^{kt}(1 - 2^{t-s})$

Suppose not. Then $n_k \le 2^{kt}(1 - 2^{t-s})$ $\forall k$

$$\sum n_k 2^{-ks} \le \sum_k (2^{t-s})^k (1 - 2^{t-s})$$
$$\le \frac{1}{(1 - 2^{t-s})}(1 - 2^{t-s}) = 1,$$

contradicts (*).

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$\therefore$ $n_k > 2^{kt}(1 - 2^{t-s})$ for some $k \in \mathbb{N}$

These n_k balls have radii $> 2^{-k-2}$,
so we can choose balls of radius 2^{-k-2}
contained inside these n_k balls with centres in F.
If $N_\delta(F)$ is the largest number of disjoint
balls of radius δ with centres in F, then

$$N_{2^{-k-2}}(F) \geq n_k > 2^{kt}(1 - 2^{t-s})$$

$$\Rightarrow N_{2^{-k-2}}(F)\left(2^{-k-2}\right)^t > 2^{-2t}(1 - 2^{t-s}) > 0$$

$$\Rightarrow \limsup_{\delta \to 0} N_\delta(F)\,\delta^t > 0$$

$\Rightarrow$ $\overline{\dim}_B F \geq t$ for any $t < \dim_p F$.

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$\therefore$ $\overline{\dim}_B F \geq \dim_p F$.

<u>Theorem</u>: $\dim_p F = \overline{\dim}_{MB} F$.

<u>Proof</u>: If $F \subseteq \bigcup_{i=1}^{\infty} F_i$, where each F_i is nonempty & bounded, then

$$\dim_p F \leq \dim_p\left(\bigcup_{i=1}^{\infty} F_i\right) = \sup_{i \in \mathbb{N}} \dim_p F_i \leq \sup_{i \in \mathbb{N}} \overline{\dim}_B F_i$$

Taking inf over all such covers $\{F_i\}$,
we get $\dim_p F \leq \overline{\dim}_{MB} F$.

To prove $\overline{\dim}_{MB} F \leq \dim_p F$, we'll show
that for $s > \dim_p F$, $\overline{\dim}_{MB} F \leq s$.

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$s > \dim_P F \Rightarrow P^s(F) = 0$

$\therefore \exists \{F_i\}_{i=1}^{\infty}$ with $F \subseteq \bigcup_{i=1}^{\infty} F_i$ such that

$P_0^s(F_i) < \infty \quad \forall i$

$\Rightarrow$ For each i, $P_\delta^s(F_i) < \infty$ for small enough $\delta > 0$.

$\Rightarrow \quad N_\delta(F_i)\, \delta^s \le C$ for some constant C for all small δ,

where $N_\delta(F_i)$ is the largest no. of disjoint balls of radius δ with centres in F_i.

$\Rightarrow \quad \overline{\dim}_B F_i \le s \quad \forall i$

$\Rightarrow \quad \overline{\dim}_{MB} F \le s$

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